Evaluating the robustness of modular ocr-llm pipelines vs. vision-language models against text degradation in bangla document understanding

Sirajum Munira
24241290

Musfikur Rahman Mahin
23201432

Rakatul Faria
22201399

***Abstract*—The digitization and understanding of Bangla documents remain challenging when document images suffer from degradation such as blur, fading, stains, occlusion, and geometric distortion. The standard system follows a modular pipeline in which an Optical Character Recognition (OCR) system first extracts text from the document image. Then a Large Language Model (LLM) understands or reconstructs the writing. The errors made in the OCR phase can still impact the results of the LLM. In contrast, recent Vision-Language Models (VLMs) can directly process document images and are able to integrate both visual and linguistic information. Thus reducing the dependence on intermediate OCR output. These two approaches are yet to be extensively studied, especially in controlled real-world document degradation scenarios for handwritten Bangla documents. This study investigates the robustness of modular OCR-LLM pipelines and VLMs for Bangla document reconstruction on a custom dataset made with Bangla handwriting images of different types of physical degradation.**

I. Introduction

The digitalization of documentation is essential for preserving information, sharing information and analyzing the textual context. However, the performance of document digitization systems can significantly decrease when documents are captured under poor visual conditions or experience physical deterioration. Real-world documents may contain blur, faded ink, stains, occlusion, noise, geometric distortion and other forms of degradation. These challenges become more significant for handwritten documents, where variations in writing style and character structure already make text recognition difficult. Bangla document understanding presents additional challenges because of the structural complexity of the Bangla script. While several OCR systems and benchmarks have been developed for Bangla text, the robustness of document understanding systems under controlled degradation conditions remains comparatively underexplored, particularly for handwritten documents.

In the traditional document understanding system, first the Optical Character Recognition (OCR) is used to convert the document into machine readable text. Then a Large Language Model (LLM) is used to understand, reconstruct or analyze the text. This pipeline is very flexible as different models can be used independently. However, this introduces a critical problem, error occurred during the OCR part may affect the overall performance. Recent developments in Vision-Language Models (VLMs) provide an alternative solution. Instead of depending on OCR models to extract text from images, VLMs can directly process visual document information. VLMs can jointly utilize visual features and linguistic context. However, the performance of VLMs may also decrease when important visual features are obscured by severe degradation.

Recent research has improved both OCR-LLM and VLMs a lot, now the question remains: which approach is more robust when Bangla handwritten documents face different forms of physical degradation? Existing evaluations often focus on clean benchmark datasets and individual recognition tasks. These do not represent a real life document digitization.

Another limitation of OCR evaluation is that character-level or word-level recognition accuracy alone may not fully represent how accurate the reconstructed document is. A system may correctly recognize a large proportion of individual characters while still producing text with missing words, incorrect ordering or structural inconsistencies.

To address this problem, this study investigates the robustness of modular OCR-LLM pipelines and Vision-Language Models for degraded Bangla handwritten document reconstruction using a custom dataset. This controlled dataset enables the systematic examination of how different degradation conditions influence the performance of both modular and end-to-end approaches.

The objectives of this research are:

- Building a custom dataset Bangla handwriting with physical degradation.
- Compare OCR-LLM pipelines and VLMs for Bangla document understanding.
- Evaluate their performance on both normal and degraded documents.
- Measure how different types and levels of degradation affect their performance.
- Identify which approach is more accurate and robust under different document conditions.

II. Literature review

To better understand the current research landscape and identify the remaining challenges, the following literature review examines previous work in four main areas: OCR systems for Bangla documents, OCR-free and Vision-Language approaches to document understanding, the effects of docu-

ment degradation and OCR errors on downstream tasks, and existing benchmarks for Bangla document understanding.

A. TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models

Li et al. address a limitation in conventional text recognition systems. These systems use a CNN encoder with an RNN decoder and rely on an external language model for post-processing. TrOCR replaces this with a single Transformer encoder-decoder model. The encoder is initialized from DeiT or BEiT. The decoder is initialized from RoBERTa or MiniLM. It generates wordpiece tokens directly, with no convolutional backbone and no external language model. Pre-training runs in two stages, first on 684M synthetic printed textlines, then on task-specific printed and handwritten data. TrOCR-Large reaches an F1 of 96.58 on SROIE, against 54.57 for Tesseract and 36.09 for CRNN. On IAM it achieves a CER of 2.89 without an external LM. The model performs recognition only, text detection is excluded and cropped textline images are required as input.

B. bbOCR: An Open-source Multi-domain OCR Pipeline for Bengali Documents

The paper addresses the limited availability of comprehensive and open-source OCR systems designed specifically for Bengali documents. This paper is a modular end to end Bengali document OCR pipeline covering geometric correction , illumination correction , document layout analysis, line and word detection and word recognition. HTML reconstruction converts layout outputs into a searchable structured format. A series of thorough experiments were conducted on BCD3 to evaluate bbOCR against Tesseract, revealing an average reduction of 8.44% in WER and 11.77% in CER for bbOCR across all domains. The pipeline handles only printed text, handwritten documents are excluded.

C. OCRBench v2: An Improved Benchmark for Evaluating Large Multimodal Models on Visual Text Localization and Reasoning

This paper. address the saturation of existing OCR benchmarks for large multimodal models. Recent models score above 96% on DocVQA, so these benchmarks can no longer separate strong models from weak ones. OCRBench v2 is a bilingual benchmark covering English and Chinese. It defines 23 tasks grouped under eight core abilities, including text recognition, text referring, text spotting, element parsing and knowledge reasoning. The data spans 31 scenarios and contains 10,000 human-verified question-answer pairs drawn from 81 screened datasets. Most models score below 50 out of 100. InternVL3-14B leads the English public set with 52.6, ahead of Gemini-1.5-Pro at 51.9. GPT-4o scores 0 at text spotting. Rotating DocVQA images dropped InternVL3-14B from 90.9% to 35.2%.

D. BANGLAWILD: An In-the-Wild Bengali Scene Text Recognition Benchmark for OCR and Vision-Language Models

Shiper et al. focuses on the lack of measurement for Bengali scene text in uncontrolled conditions. Existing resources cover handwritten documents or signboard parsing but never test OCR and VLMs on the same data. BanglaWild contains 2,535 images photographed on walls, signboards, vehicles, banners and packaging. Each image has a verbatim transcription that preserves in-image spelling errors. A separate standard spelling is stored for the 3.43% of images with non-standard text. Images also carry category labels and four diagnostic attributes. Fifteen VLMs and three OCR systems are tested under three prompts and six open models are fine-tuned with LoRA. Gemini 2.5 Flash leads with CER 14.08 and WER 29.97. EasyOCR is the best conventional system at CER 48.43 and Tesseract reaches 98.46. Visual misreading causes about 60% of errors and conjunct errors under 2%. Code-mixed Bengali-English images were excluded at filtering.

E. Enhancement of Bengali OCR by Specialized Models and Advanced Techniques for Diverse Document Types

Rabby et al. investigate the problem of building a more capable Bengali OCR system for diverse document types. This is a Bengali OCR pipeline using a self-attentional VGG-based multi-headed neural network for character recognition, specialized word segmentation models per document type (computer-composed, letterpress, typewriter, handwritten) module. The proposed system goes beyond simple text recognition. It includes document layout reconstruction and attempts to preserve structural properties. Specialized word-segmentation approaches are used for different document categories, while the architecture is designed to support both static and dynamic handwritten inputs. A queuing mechanism is additionally introduced to improve scalability and processing efficiency. On held-out documents per type, the system achieves an average Levenshtein distance-based accuracy of 87.20% and confusion matrix-based accuracy of 98.05%, versus Tesseract's 57.56% average. Evaluation uses an internal proprietary test set with no public release. The system addresses four document types but does not generalize to mixed domain pages.

F. BornoDrishti: Leveraging Vision Encoders and Domain-Adaptive Learning for Bangla OCR on Diverse Documents

BornoDrishti focuses on improving Bangla OCR across diverse document domains by leveraging modern vision encoders and domain-adaptive learning. It's built on ViT-B/16, trained via CLIP-style contrastive learning. The work uses modern visual representation learning to extract features from document images and incorporates domain-adaptive strategies to improve generalization. Instead of treating each document category as completely independent, the approach seeks to exploit transferable visual knowledge while reducing the negative effects of distributional differences. On a merged printed/handwritten word dataset, BornoDrishti achieves 83.77% accuracy with progressive learning versus 77.76% without. In single-domain settings it reaches

94.7% (printed) and 68.9% (handwritten), surpassing Tesseract (73.49% / 35.80%). Handwritten accuracy lags 26 points behind printed, constrained by limited writer diversity (24 writers) in training data.

G. When Good OCR Is Not Enough: Benchmarking OCR Robustness for Retrieval-Augmented Generation

Sun et al. examine an important weakness in evaluating OCR systems: high OCR accuracy does not necessarily produce strong downstream language-model performance. The paper focuses on industrial retrieval-augmented generation (RAG) systems, where OCR converts visual documents into textual information before retrieval and generations take place. The authors introduce an OCR benchmark containing 11 challenging document categories. These include documents with extreme layouts, high-resolution pages, complex or watermarked backgrounds, historical documents with unusual reading orders, visually decorated text, tables and mathematical formulas. In InduOCRBench annotations capture text content, logical structure and visual attributes. Ten OCR models across four paradigms are evaluated end to end through a fixed RAG pipeline using RAGAS metrics. All models degrade substantially on InduOCRBench relative to OmniDocBench. PaddleOCR-VL drops from 92.9% to 78.2%; GPT-4o from 75.0% to 52.0%. The benchmark covers OCR-first pipelines only, not native multimodal RAG architectures. The gap between character-level accuracy and downstream retrieval utility remains an open architectural challenge.

H. OCR-Free Document Understanding Transformer

Kim et al. propose Donut (Document Understanding Transformer), an influential approach to visual document understanding that eliminates the need for a separate OCR engine. Traditional document-understanding pipelines usually first apply OCR to extract text and then pass the recognized text, often together with layout information, to a downstream model. A Transformer based model processes the document image directly and generates task-specific textual outputs. Instead of depending on externally recognized text, the model learns visual and linguistic information jointly. The authors also provide a synthetic document data generator for pretraining, enabling the model to support multiple languages and domains more flexibly. Experimental results demonstrate competitive or state-of-the-art performance on several visual document understanding tasks while also showing advantages in efficiency. Performance on DocVQA trails the best OCR-based model.

I. Pix2Struct: Screenshot Parsing as Pretraining for Visual Language Understanding

Pix2Struct is a general-purpose visual-language model designed to understand visually situated language, including documents, webpages, user interfaces, illustrations, tables and natural images. Pix2Struct is an image-encoder and text-decoder model. Its main contribution is a novel pretraining strategy based on masked screenshot parsing. The model learns to reconstruct simplified HTML representations from partially masked webpage screenshots. This objective provides large-scale training data because webpages contain extensive visual and structural information represented in HTML. Pix2Struct also uses a variable-resolution input representation, allowing the model to process images with varying sizes and supports visual question answering by rendering textual prompts directly onto the input image. Pix2Struct achieves state-of-the-art on 6 of 9 benchmarks across four domains. Performance on text-dense document benchmarks trails OCR-pipeline methods that use dedicated text encoders.

J. Nougat: Neural Optical Understanding for Academic Documents

Nougat, Neural Optical Understanding for Academic Documents, proposes an end to end neural approach for converting academic document images into a markup language. Academic papers are particularly difficult to process because they contain mathematical equations, tables, figures, citations, multi-column layouts and other complex structural elements. A Swin Transformer encoder encodes page images and an mBART decoder generates tokens. Training pairs are constructed by compiling arXiv LaTeX sources. Data augmentation simulates scanned document degradation. It performs more than literal character recognition. It learns to interpret the visual structure and semantic representation of academic documents and produce a useful machine-readable output. On the arXiv test set, Nougat base achieved an edit distance of 0.071, BLEU of 89.1 and F1 of 93.1 across all text modalities. The model degenerates into repetitive loops on out-of-domain documents, this occurs in 1.5% of in-domain pages but more frequently otherwise.

K. BaFCo: A Document Understanding Benchmark for Complex Bangla Form Comprehension

BaFCo introduces a benchmark specifically designed for Bangla document and form comprehension, addressing the shortage of high-quality evaluation resources for low-resource languages. The benchmark focuses on two important document-understanding tasks: Document Layout Analysis (DLA) and Key Information Extraction (KIE). It defines a 26 type fine-grained annotation schema and a 5 type coarse schema, with entity relationships. Five flagship MLLMs are evaluated under zero-shot and CoT prompts at two reasoning effort levels. The study evaluates modern multimodal large language models from the ChatGPT, Gemini, Claude, Qwen and Kimi families using zero-shot and chain-of-thought prompting under different reasoning settings. Gemini 3 Pro achieves the best DLA results: mAP of 0.1177 (granular) and 0.2646 (coarse) at IoU@0.3 under high reasoning. DLA performance remains low overall.

L. Evaluating Large Vision Language Models on Bangla Medical Visual Question Answering

Ahmed et al. evaluate the ability of large vision-language models (LVLMs) to perform visual question answering in Bangla, focusing on the specialized medical domain. Nine

LVLMs are evaluated under zero-shot, Chain-of-Thought and LoRA fine tuning strategies, covering both open-source and closed-source models. This experimental design makes it possible to examine both the out of the box capabilities of general-purpose models and the benefits of adaptation.

Gemini 2.5 Flash achieves the highest zero-shot LAVE score of 55.44%. After LoRA fine-tuning, Med-Gemma 4B and Qwen2.5-VL reach 69.95% and 67.18% LAVE respectively. Fine-tuning improves overall performance, yet important weaknesses remain in specialized reasoning. Expert validation was conducted on only 1,000 of 7,000 pairs due to resource constraints.

M. Lost in OCR Translation? Vision-Based Approaches to Robust Document Retrieval

This paper examines whether vision-based retrieval is more robust than OCR for RAG on scanned documents. It compares OCR-based and VLM-based RAG pipelines. Existing benchmarks use clean pages and weak OCR baselines such as Tesseract. The authors build DocDeg, a set of 4,196 documents from the OSTI archive. Two annotators label each page into four degradation levels, from native digital to severely degraded. Ten question-answer pairs per document are generated with Llama 3.3 70B. The vision pipeline is ColQwen2 7B, which embeds image patches and scores them with ColBERT late interaction. The OCR pipeline extracts text with Llama 3.2 90B or Nougat, then embeds it with Qwen2. The OCR pipeline wins at every degradation level, with overall MRR 0.4852 against 0.2471 for the vision pipeline. The vision pipeline uses far less memory, 1.38 GB per 1,000 documents against 9.5 GB. Neither pipeline was fine-tuned on DocDeg.

N. Evaluating Robustness of LLMs in Question Answering on Multilingual Noisy OCR Data

Piryani et al. examine how OCR noise affects question answering on historical documents. Prior work studied OCR errors in retrieval and entity recognition but not in QA. MultiOCR-QA is built from the ICDAR 2019 Post-OCR Correction dataset in English, French and German. Every passage appears twice, as RawOCR text and as corrected ground truth. QA pairs are generated by a LLaMA-3.1-70B model fine-tuned per language. Paragraphs are sorted into low, medium and high noise bands by character error rate. The final set holds 50,079 pairs. Six LLMs are tested with BERTScore, Exact Match and a Contains metric. Every model drops on raw text. Gemma-2 27B is strongest throughout. German suffers most. The dataset covers no low-resource languages or scripts.

O. Introducing Sarvam Akshar

The paper compares native 3-billion-parameter Vision-Language Models with traditional OCR systems across 22 Indic languages, including Bengali. Instead of breaking documents into smaller parts, the system processes the entire page image at once, which helps preserve important features such as the layout, matras, and conjunct letters. It also uses a verification step to check the results. The system performs better than older tools such as Tesseract and Google Cloud Vision, especially when reading multi-column documents and historical texts, with fewer mistakes in characters and words. However, processing high-resolution images requires a lot of computer memory, and the model may sometimes produce incorrect spellings when the text is very unclear or damaged. The study also does not test the system thoroughly on specific types of image problems, such as motion blur, fading, and noise, or examine how it could be optimized to run on smaller devices.

P. BN-HTRd: A Benchmark Dataset for Document Level Offline Bangla Handwritten Text Recognition (HTR) and Line Segmentation

Rahman et al. present a method that automatically separates individual lines of text in handwritten Bangla documents without needing labeled training data. The method uses image-processing techniques such as morphological operations, Hough transforms, and clustering to identify curved text lines. When tested on their BN-HTRd dataset, which contains 786 handwritten pages, the system achieved an average F-measure of 81.57% and mAP of 0.547. However, it has difficulty when text lines overlap or when the document is heavily tilted. It also does not work well when printed and handwritten text appear together. The study does not test newer AI-based Vision-Language Models or OCR-LLM systems for reconstructing complete pages, and it also does not examine how well the method works with real-world document damage such as aging paper, faded ink, or low-quality camera images.

Q. Lost in OCR Translation? Vision-Based Approaches to Robust Document Retrieval

Most et al. study whether vision-based models, such as ColQwen2, can handle damaged documents better than traditional OCR-based systems using Llama 3.2 90B and Nougat. They test both types of systems on four levels of document damage using their DocDeg benchmark. The results show that OCR-based systems perform better than vision-based models at all levels of damage, achieving an MRR of 0.4852 compared to 0.2471. However, vision-based models need much less memory for storing and searching documents, using 1.38 GB compared to 9.5 GB for every 1,000 documents. The study only tests the models without fine-tuning and focuses on English administrative documents. It also does not examine how these systems perform on complex non-Latin scripts such as Bangla, which have unique features like matras and conjunct letters, or whether fine-tuning VLMs could improve their accuracy.

R. How Robust is OCR-Reasoning? Evaluating OCR-Reasoning Robustness of Vision-Language Models under Visual Perturbations

Cheng et al. introduce OCR-Robust, a benchmark with 812 visual question-answering samples covering documents, scene text, and structured graphics. The benchmark tests how five types of visual damage at three different severity levels affect

OCR-based reasoning. They evaluate 18 models, including commercial VLMs, open-source VLMs, and OCR+LLM systems. The results show that closed-source models are more stable when image quality decreases, while high accuracy on clear images does not always mean that a model will perform well on damaged images. They also find that charts and tables are more easily affected by visual damage than normal document layouts. However, the study mainly focuses on short question-answering tasks rather than reconstructing complete pages of text. It also only tests English and does not examine low-resource non-Latin scripts such as Bangla, which contain complex features like mantras and stacked conjunct letters, under different types of image degradation.

III. Dataset

To evaluate the robustness of OCR-LLM pipelines and Vision-Language Models on degraded Bangla handwritten documents, a custom dataset was created. The dataset will consist of handwritten Bangla documents collected from multiple writers and photographed under controlled conditions. Each document will be captured after physically degrading it using different types of damage, such as blur, fading, stains, occlusion, distortion or other real-world degradation conditions. Each image will be paired with manually verified ground-truth text representing the original handwritten content. The dataset will allow a controlled evaluation of how different types of physical document degradation affect the performance of OCR-LLM pipelines and Vision-Language Models in Bangla document reconstruction.

References

[1] Li, M., Lv, T., Chen, J., Cui, L., Lu, Y., Florencio, D., Zhang, C., Li, Z., & Wei, F. (2023). TrOCR: Transformer-based optical character recognition with pre-trained models. Proceedings of the AAAI Conference on Artificial Intelligence, 37(11), 13094–13102. https://doi.org/10.1609/aaai.v37i11.26538

[2] Zulkarnain, I. M., et al. (2023). bbOCR: An open-source multi-domain OCR pipeline for Bengali documents. arXiv.

[3] Fu, L., Kuang, Z., Song, J., Huang, M., Yang, B., Li, Y., Zhu, L., Luo, Q., Wang, X., Lu, H., Li, Z., Tang, G., Shan, B., Lin, C., Liu, Q., Wu, B., Feng, H., Liu, H., Huang, C., ... Bai, X. (2025). OCRBench v2: An improved benchmark for evaluating large multimodal models on visual text localization and reasoning. Advances in Neural Information Processing Systems 38: Datasets and Benchmarks Track.

[4] Shiper, S., Dipto, T. T., Inzamam, M. M., & Tanzeem, E. (2026). BanglaWild: An in-the-wild Bengali scene text recognition benchmark for OCR and vision-language models (arXiv:2608.03884). arXiv. https://arxiv.org/abs/2608.03884

[5] Rabby, A. S. A., Ali, H., Islam, M. M., Abujar, S., & Rahman, F. (2024). Enhancement of Bengali OCR by specialized models and advanced techniques for diverse document types. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (pp. 1102–1109).

[6] Islam, S. M. J., et al. (2026). BornoDrishti: Leveraging vision encoders and domain-adaptive learning for Bangla OCR on diverse documents. In Proceedings of the EACL 2026 Industry Track. Association for Computational Linguistics.

[7] Sun, L., Wangdexian, Huang, J., Zhang, L., Jia, C., Cheng, Z., & Zhang, X. (2026). When good OCR is not enough: Benchmarking OCR robustness for retrieval-augmented generation. In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track) (pp. 884–894). Association for Computational Linguistics. https://doi.org/10.18653/v1/2026.acl-industry.60

[8] Kim, G., Hong, T., Yim, M., Nam, J., Park, J., Yim, J., Hwang, W., Yun, S., Han, D., & Park, S. (2022). OCR-free document understanding transformer. In Computer Vision – ECCV 2022. Springer.

[9] Lee, K., Joshi, M., Turc, I. R., Hu, H., Liu, F., Eisenschlos, J. M., Khandelwal, U., Shaw, P., Chang, M.-W., & Toutanova, K. (2023). Pix2Struct: Screenshot parsing as pretraining for visual language understanding. In Proceedings of the 40th International Conference on Machine Learning (Vol. 202, pp. 18893–18912). PMLR.

[10] Blecher, L., Cucurull, G., Scialom, T., & Stojnic, R. (2024). Nougat: Neural optical understanding for academic documents. In The Twelfth International Conference on Learning Representations

[11] Azad, A. T., Apan, I. S., Ahmed, F., Katha, S. K., Jubaer, E., Alam, A., Nandi, P. K., Ali, A. A., Chadha, A., Islam, M. M., & Rahman, A. M. (2026). BaFCo: A document understanding benchmark for complex Bangla form comprehension [Preprint]. arXiv. .

[12] Ahmed, R., Tahmid, I., Hossain, M. S., Tomal, T. H., Jawad, M. M., Uddin, A. B., Fahim, M., & Bhuiyan, M. F. A. (2026). Evaluating large vision language models on Bangla medical visual question answering. In Findings of the Association for Computational Linguistics: ACL 2026 (pp. 37362–37378). Association for Computational Linguistics. https://doi.org/10.18653/v1/2026.findings-acl.1862

[13] Most, A., Winjum, J., Biswas, A., Jones, S., Ranasinghe, N. R., O'Malley, D., & Bhattarai, M. (2025). Lost in OCR translation? Vision-based approaches to robust document retrieval. In Proceedings of the 2025 ACM Symposium on Document Engineering (pp. 1–10). ACM. https://doi.org/10.1145/3704268.3742698

[14] Piryani, B., Mozafari, J., Abdallah, A., Doucet, A., & Jatowt, A. (2025). Evaluating robustness of LLMs in question answering on multilingual noisy OCR data. In Proceedings of the 34th ACM International Conference on Information and Knowledge Management. ACM. https://doi.org/10.1145/3746252.3761295